

Matrices

Matrices and Their Significance

Matrices are fundamental mathematical structures used to represent and manipulate data in various fields, particularly in linear algebra. They are rectangular arrays of numbers, symbols, or expressions arranged in rows and columns. The significance of matrices lies in their ability to represent linear transformations, solve systems of equations, and perform operations in various applications such as computer graphics, statistics, and engineering.

Matrix Notation

Matrices are typically denoted by uppercase letters (e.g., A,B,C), while their elements are represented by lowercase letters with two subscripts indicating the row and column (e.g., a_{ij} for the element in the i -th row and j -th column). Matrices can be enclosed in square brackets or parentheses. For example, a matrix A of size $m \times n$ can be represented as:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Dimension (Order) of a Matrix

The dimension or order of a matrix is defined by the number of rows and columns it contains. If a matrix has m rows and n columns, it is said to be of order $m \times n$. For example, a matrix with 3 rows and 4 columns is a 3×4 matrix. The total number of elements in the matrix is given by the product mn ^[2]

Types of Matrix

Singleton Matrix

A matrix that has only one element is called a singleton matrix. In this type of matrix number of columns and the number of rows is equal to 1. A singleton matrix is represented as $[a]_{1 \times 1}$.

Example of Singleton Matrix

$$[5]_{1 \times 1}$$

In the above example of the Singleton Matrix, there is only one element 5. Hence there is only one column and only one row.

Null Matrix

A matrix whose all elements are zero is called a Null Matrix. A null matrix is also called a Zero Matrix because its all elements are zero. An example of a null matrix is mentioned below:

Example of Zero Matrix

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}$$

In the above example of a null or zero matrix, all the elements are zero. Hence the given example is a matrix of order 3×3 whose all elements are zero.

Row Matrix

A matrix that contains only one row and any number of columns is known as a row matrix. A row matrix is represented as $[a]_{1 \times n}$ where 1 is the number of row and n is the number of columns present in a row matrix. An example of a row matrix is given below

Example of Row Matrix

$$[1 \ 3 \ 7]_{1 \times 3}$$

In the above example of a row matrix, the number of rows is 1, and the number of columns is 3. Hence the order of the matrix is 1×3 .

Column Matrix

A matrix that contains only one column and any number of rows is called a Column Matrix. A Column Matrix is represented as $[a]_{n \times 1}$ where n is the number of rows and 1 is the number of columns. An example of a column matrix is given below:

Example of Column Matrix

$$\begin{bmatrix} 1 \\ 15 \\ 4 \\ 5 \end{bmatrix}_{4 \times 1}$$

In the above example of a column matrix the number of rows is 4 and the number of columns is 1 thus making it a matrix of order 4×1 .

Horizontal Matrix

A matrix in which the number of rows is lower than the number of columns is called a Horizontal Matrix. columns

Example of Horizontal Matrix

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}_{2 \times 4}$$

In the above matrix, the number of rows is 2 while the number of columns is 4 thus making it a horizontal matrix.

Vertical Matrix

The matrix in which the number of rows exceeds the number of columns is called a Vertical Matrix. A Vertical matrix is represented as $[a]_{i \times j}$ where $i > j$. An example of a Vertical Matrix is mentioned below:

Example of Vertical Matrix

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \\ 7 & 8 \end{bmatrix}_{4 \times 2}$$

In the above matrix, the number of rows is 4 while the number of columns is 2 thus making it a Vertical matrix.

Rectangular Matrix

A matrix that does not have an equal number of rows and columns is known as a Rectangular Matrix. A rectangular matrix can be represented as $[A]_{m \times n}$ where $m \neq n$. An example of a rectangular matrix is mentioned below:

Example of Rectangular Matrix

$$\begin{bmatrix} 1 & 3 & 7 & 15 \\ 3 & 4 & 6 & 11 \\ 5 & 2 & 9 & 8 \end{bmatrix}_{3 \times 4}$$

In the above example, we see that the number of rows is 3 while the number of columns is 4 i.e. both are unequal thus making it a rectangular matrix. We can say that both horizontal and vertical [matrices](#) are examples of rectangular matrices.

Square Matrix

A matrix that has an equal number of rows and an equal number of columns is called a Square Matrix. Generally, the representation used for the square matrix is $[A]_{n \times n}$. An example of Square Matrix is mentioned below:

Example of Square Matrix

$$\begin{bmatrix} 8 & 3 & 2 \\ 6 & 4 & 6 \\ 5 & 7 & 9 \end{bmatrix}_{3 \times 3}$$

In the above example of Square Matrix, both the number of rows and columns are 3, thus making them seem like a square structure.



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Diagonal Matrix

A matrix that has all elements as 0 except diagonal elements is known as a diagonal matrix. A Diagonal Matrix is only possible in the case of a Square Matrix. An example of a Diagonal Matrix is mentioned below:

Example of Diagonal Matrix

$$\begin{bmatrix} 8 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 9 \end{bmatrix}_{3 \times 3}$$

In the above example, the diagonal elements are 8, 4, and 9 and the rest elements are zero.

Scalar Matrix

A diagonal matrix whose all diagonal elements are non-zero and the same is called a Scalar Matrix. Scalar Matrix is a kind of diagonal matrix where all diagonal elements are the same. Identity Matrix is a special case of Scalar Matrix.

Example of Scalar Matrix

$$\begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}_{3 \times 3}$$

In the above example, the given matrix is a diagonal matrix whose all diagonal elements are 4, and hence, this is an example of a Scalar Matrix.

Identity Matrix

A diagonal matrix where all the diagonal elements are 1 and all non-diagonal elements are 0 is called an Identity Matrix. The Identity Matrix is called Unit Matrix. The identity matrix or unit matrix always has an equal number of rows and columns.

Example of Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3}$$

In the above diagonal matrix of order 3×3 , all the diagonal elements are 1, and non-diagonal elements are zero. Hence this diagonal matrix is an Identity Matrix.

Triangular Matrix

A square matrix in which the non-zero elements form a triangular below and above the diagonal is called a Triangular Matrix. Based on the triangle formed below or above the diagonal, the triangular matrix is classified as:

- Upper Triangular Matrix
- Lower Triangular Matrix

Let's learn them in detail.

Upper Triangular Matrix

A square matrix in which all the elements below the diagonal are zero and the elements from the diagonal and above are non-zero elements is called an Upper Triangular Matrix. In an Upper Triangular Matrix, the non-zero elements form a triangular-like shape.

Example of Upper Triangular Matrix

$$\begin{bmatrix} 8 & 5 & 6 \\ 0 & 4 & 7 \\ 0 & 0 & 9 \end{bmatrix}_{3 \times 3}$$

In the above example of the Upper Triangular Matrix, all the elements below the diagonal are zero.

Lower Triangular Matrix

A square matrix in which all the elements above the diagonal are zero and the elements from the diagonal and below are non-zero elements is called a Lower Triangular Matrix. In a Lower Triangular Matrix, the non-zero elements form a triangular-like shape from the diagonal and below.

Example of Lower Triangular Matrix

$$\begin{bmatrix} 8 & 0 & 0 \\ 6 & 4 & 0 \\ 5 & 7 & 9 \end{bmatrix}_{3 \times 3}$$

In the above example of the lower triangular matrix, all the elements above the diagonal are zero.

Singular Matrix

A singular matrix is referred to as a square matrix whose determinant is zero and is not invertible. If $\det A = 0$, a square matrix "A" is said to be singular; otherwise, it is said to be non-singular.

Example of Singular Matrix

$$A = \begin{bmatrix} 3 & 6 & 9 \\ 6 & 12 & 18 \\ 2 & 4 & 6 \end{bmatrix}$$

$$\Rightarrow |A| = 3(12 \times 6 - 18 \times 4) - 6(6 \times 6 - 18 \times 2) + 9(6 \times 4 - 12 \times 2)$$

$$\Rightarrow |A| = 3(72 - 72) - 6(36 - 36) + 9(24 - 24)$$

$$\Rightarrow |A| = 3 \times 0 - 6 \times 0 + 9 \times 0 = 0$$

Non Singular Matrix

A Non-Singular matrix is defined as a square matrix whose determinant is not equal to zero and is invertible.

Example of a Non-Singular Matrix

$$|A| = \begin{bmatrix} 1 & 5 \\ 9 & 8 \end{bmatrix}$$

$$\Rightarrow |A| = 8 \times 1 - 9 \times 5 = 8 - 45 = -37$$

Symmetric Matrix

A square matrix “A” of any order is defined as a symmetric matrix if the transpose of the matrix is equal to the original matrix itself, i.e., $A^T = A$.

Example of Symmetric Matrix

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Skew Symmetric Matrix

A square matrix “A” of any order is defined as a skew-symmetric matrix if the transpose of the matrix is equal to the negative of the original matrix itself, i.e., $A^T = -A$.

Example of Skew Symmetric Matrix

$$\begin{bmatrix} 0 & 3 & 5 \\ -3 & 0 & -2 \\ -5 & 2 & 0 \end{bmatrix}$$

Orthogonal Matrix

A square matrix whose transpose is equal to its inverse is called Orthogonal Matrix. In an Orthogonal Matrix if $A^T = A^{-1}$ then $AA^T = I$ where I is the Identity Matrix.

Example of Orthogonal Matrix

$$A = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$\text{and } A^T = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$\Rightarrow A \times A^T = \begin{bmatrix} \cos^2(\theta) + \sin^2(\theta) & \cos(\theta)\sin(\theta) - \cos(\theta)\sin(\theta) \\ \sin(\theta)\cos(\theta) - \cos(\theta)\sin(\theta) & \cos^2(\theta) + \sin^2(\theta) \end{bmatrix}$$

$$\Rightarrow A \times A^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_{(2 \times 2)}$$

Idempotent Matrix

An idempotent matrix is a special type of square matrix that remains unchanged when multiplied by itself, i.e., $A^2 = A$.

Example of Idempotent Matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\text{Hence, } A \cdot A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = A$$

Nilpotent Matrix

A Nilpotent is a square matrix that when raised to some positive power results in zero matrix. The least power let's say 'p' for which the matrix yields zero matrix, then it is called the Nilpotent Matrix of power 'p'.

Example of Nilpotent Matrix

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow A^2 = \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{and } A^3 = A \cdot A^2$$

$$\Rightarrow A^3 = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, A is a Nilpotent Matrix of index 3.

Periodic Matrix

A periodic matrix is a square matrix which exhibits periodicity, i.e. when raised to some power let's say p+1 then $A^{p+1} = A$. If p = 1 then $A^2 = A$ it means A is an Idempotent Matrix. Thus we can say that Idempotent Matrix is a case of Periodic Matrix.

Example of Periodic Matrix

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

The above square matrix is a Periodic Matrix of Period 2, where p = 1.

Involuntary Matrix

An involuntary matrix is a special type of square matrix whose inverse is the original matrix itself, i.e., $P = P^{-1}$, or, in other words, its square is equal to an identity matrix i.e. $P^2 = I$.

Example of Involuntary Matrix

$$A = \begin{bmatrix} 2 & 1 \\ -3 & -2 \end{bmatrix}$$

Trace of a Matrix

The sum of diagonal elements of a matrix is known as the trace of a matrix. The Trace of a matrix A can be represented as $\text{tr}(A)$. The Trace of a matrix can be calculated for a square matrix only.

Example:

$$A = \begin{bmatrix} 15 & 12 & 9 \\ 4 & 6 & 11 \\ 5 & 9 & 0 \end{bmatrix}_{3 \times 3}$$

$$\text{tr}(A) = 15 + 6 + 0 = 21$$

Hermitian Matrix

A complex square matrix is called a Hermitian Matrix if the conjugate transpose of the matrix is equal to the original matrix. In this type of matrix the diagonal elements must be a real number.

Example of Hermitian Matrix

$$A = \begin{bmatrix} 3 & 1-i \\ 1+i & 4 \end{bmatrix}$$

Skew Hermitian Matrix

A complex square matrix is called a Skew Hermitian Matrix if the conjugate transpose of the matrix is equal to the negative of the original matrix. In this type of matrix the diagonal elements can be either 0 or a complex number but can not be real number other than 0.

Example of Skew Hermitian Matrix

$$A = \begin{bmatrix} 2i & 2i & -3i \\ -2i & 0 & 4 \\ 3i & -4 & 0 \end{bmatrix}$$

Orthogonal Matrix

A Matrix is called Orthogonal Matrix when the transpose of Matrix is inverse of that matrix or the product of Matrix and it's transpose is equal to an Identity Matrix.

Mathematically, an $n \times n$ matrix A is considered orthogonal if

$$A^T = A^{-1}$$

OR

$$AA^T = A^TA = I$$

Where,

- A^T is the transpose of the square matrix,
- A^{-1} is the inverse of the square matrix, and
- I is the identity matrix of same order as A .

Condition for Orthogonal Matrix

For any matrix to be an orthogonal Matrix, it needs to fulfil the following conditions:

- Every two rows and two columns have a dot product of zero, and
- Every row and every column has a magnitude of one.

Properties of an Orthogonal Matrix

Some of the properties of Orthogonal Matrix are:

- Inverse and Transpose are equivalent. i.e., $A^{-1} = A^T$.
- An identity matrix is the outcome of A and its transpose. That is, $AA^T = A^T A = I$.
- In light of the fact that its determinant is never 0, an orthogonal matrix is always non-singular.
- An orthogonal diagonal matrix is one whose members are either 1 or -1.
- A^T is orthogonal as well. A^{-1} is also orthogonal because $A^{-1} = A^T$.
- The eigenvalues of A are ± 1 and the eigenvectors are orthogonal.
- As $I \times I = I \times I = I$, and $I^T = I$. Thus, I an identity matrix (I) is orthogonal.

Example of 2×2 Orthogonal Matrix

Let's consider the an 2×2 i.e., $A = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix}$.

Let's check this using the product of the matrix and its transpose.

$$A^T = \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$$

$$\text{Thus, } A \cdot A^T = \begin{bmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{bmatrix} \cdot \begin{bmatrix} \cos x & -\sin x \\ \sin x & \cos x \end{bmatrix}$$

$$\Rightarrow A \cdot A^T = \begin{bmatrix} \cos^2 x + \sin^2 x & \sin x \cos x - \sin x \cos x \\ \cos x \sin x - \cos x \sin x & \sin^2 x + \cos^2 x \end{bmatrix}$$

$$\Rightarrow A \cdot A^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Which is an Identity Matrix.

Thus, A is an example of an Orthogonal Matrix of order 2×2.

Determinant of Orthogonal Matrix

Determinant of any Orthogonal Matrix is either +1 or -1. Here, let's demonstrate the same. Imagine a matrix A that is orthogonal.

For any orthogonal matrix A , we know

$$A \cdot A^T = I$$

Taking determinants on both sides,

$$\det(A \cdot A^T) = \det(I)$$

$$\Rightarrow \det(A) \cdot \det(A^T) = 1$$

As, determinant of identity matrix is 1 and $\det(A) = \det(A^T)$

$$\text{Thus, } \det(A) \cdot \det(A) = 1$$

$$\Rightarrow [\det(A)]^2 = 1$$

$$\Rightarrow \det(A) = \pm 1$$

Inverse of Orthogonal Matrix

The inverse of the orthogonal matrix is also orthogonal as inverse is same transpose for orthogonal matrix. As for any matrix to be an orthogonal, inverse of the matrix is equal to its transpose.

For an Orthogonal matrix, we know that

$$A^{-1} = A^T$$

$$\text{Also } A \cdot A^T = A^T \cdot A = I \dots (i)$$

Let two matrix A and B and if they are inverse of each other then,

$$A \cdot B = B \cdot A = I \dots (ii)$$

From (i) and (ii),

$$B = A^T \text{ which is same as } A = A^T$$

So, we conclude that the transpose of an orthogonal matrix is its inverse only.

Hermitian Matrix

A square matrix is said to be a Hermitian matrix if it is equal to its conjugate transpose matrix.

A complex square matrix " $A_{n \times n} = [a_{ij}]$ " is said to be a Hermitian matrix if

$$\mathbf{A} = \mathbf{A}^H$$

where A^H is the conjugate transpose of matrix A .

In other words, " $A_{n \times n} = [a_{ij}]$ " is said to be a Hermitian matrix if $a_{ij} = \bar{a}_{ji}$, where $\bar{a}_{ji}$ is the complex conjugate of a_{ji} .

Examples of Hermitian Matrix

- Matrix given below is a Hermitian matrix of order " 2×2 ."

$$A = \begin{bmatrix} 8 & 1+i \\ 1-i & 5 \end{bmatrix}$$

Now, the conjugate of $A \Rightarrow$

$$\bar{A} = \begin{bmatrix} 8 & 1-i \\ 1+i & 5 \end{bmatrix}$$

The conjugate transpose of matrix $A \Rightarrow$

$$A^H = (\bar{A})^T = \begin{bmatrix} 8 & 1+i \\ 1-i & 5 \end{bmatrix}$$

We can see that $A = A^H$, so the given matrix is a Hermitian matrix.

- Matrix given below is a Hermitian matrix of order " 3×3 ."

$$B = \begin{bmatrix} 1 & 2+3i & 4i \\ 2-3i & 0 & 6-7i \\ -4i & 6+7i & 3 \end{bmatrix}$$

Properties of Hermitian Matrix

Some important properties of Hermitian matrix are discussed below:

- Principal diagonal entries of a Hermitian matrix are always real.
- Non-diagonal entries of a Hermitian matrix are complex numbers.

- If A is a Hermitian matrix of any order and k is a real scalar, then kA is also a Hermitian matrix as $(kA)^H = kA^H = kA$.
- When two Hermitian matrices of the same order are added or subtracted, the resulting matrix is also a Hermitian matrix.
- When two Hermitian matrices are multiplied, the resultant matrix is also a Hermitian matrix, if and only if $AB = BA$.
- Trace of a Hermitian matrix is always a real number.
- Determinant of a Hermitian matrix is always a real number.
- Inverse of the Hermitian matrix is also a Hermitian matrix.
- Conjugate matrix of a Hermitian matrix is also Hermitian.
- If A is a Hermitian matrix of any order, then A^n is also a Hermitian matrix for all positive integers n .

Skew-Hermitian Matrix

A complex square matrix is said to be a skew-Hermitian matrix if the conjugate transpose matrix is equal to the negative of the original matrix. A square matrix " $A_{n \times n} = [a_{ij}]$ " is said to be a Hermitian matrix if $A^H = -A$, where A^H is the conjugate transpose of matrix A .

Matrix given below is a Hermitian matrix of order " 2×2 ."

$$A = \begin{bmatrix} ai & -b + ci \\ -b - ci & 0 \end{bmatrix}$$

Now, the conjugate of $A \Rightarrow$

$$\overline{A} = \begin{bmatrix} -ai & -b - ci \\ -b + ci & 0 \end{bmatrix}$$

The conjugate transpose of matrix $A \Rightarrow$

$$A^H = (\overline{A})^T = \begin{bmatrix} -ai & -b + ci \\ -b - ci & 0 \end{bmatrix} = -A$$

We can see that $A^H = -A$, so the given matrix is a skew-Hermitian matrix.

Symmetric Matrix

If for a matrix, the transposed form of that matrix is the same as the original matrix, then that matrix is said to be a Symmetric Matrix. Let, a square matrix A of size $n \times n$ is said to be symmetric if

$$A^t = A$$

Where,

- $[a_{ij}] = [a_{ji}]$, for $1 \leq i \leq n$, and $1 \leq j \leq n$.
- $[a_{ij}]$ is an element at position (i, j) which is i^{th} row and j^{th} column in matrix A , and
- $[a_{ji}]$ is an element at position (j, i) which is j^{th} row and i^{th} column in matrix A .

So, $[a_{ji}]$ represents the transpose of $[a_{ij}]$ matrix.

For example, let consider a square matrix $A = \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}$,

and Transpose of $A = \begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}$

So, this matrix is a Symmetric Matrix, because the transposed form of this matrix is itself the original matrix.

Properties of Symmetric Matrices

Some Properties of Symmetric Matrices are as follows:

- All [eigenvalues](#) of a symmetric matrix are real.
- Eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are orthogonal.
- A symmetric matrix is always diagonalizable (i.e., it can be written as a diagonal matrix with its eigenvalues on the diagonal).
- The sum and product of two symmetric matrices are symmetric.
- A symmetric matrix is positive definite if all its eigenvalues are positive.
- The spectral theorem states that every symmetric matrix can be diagonalized by an orthogonal matrix.

Property: If matrix A is a square matrix then $(A + A^t)$ is always symmetric.

Prove:

To find if a matrix symmetric or not, first, we have to find the transposed form of the given matrix

So, let's find the transpose of $(A + A^t)$

$$= (A + A^t)^t$$

$$= A^t + (A^t)^t$$

$$= A^t + A \quad [\text{here, } (A^t)^t = A]$$

$$= (A + A^t)$$

So, this is the same as the given matrix, so it is symmetric.

Skew Symmetric Matrix

If for a matrix, the transposed form of that matrix is the same as the negative of the original matrix, then that matrix is said to be a Skew-Symmetric Matrix. Let, a square matrix A of size $n \times n$ is said to be skew-symmetric if

$$A^t = -A$$

Where,

- $[a_{ij}] = -[a_{ji}]$, for $1 \leq i \leq n$, and $1 \leq j \leq n$.
- $[a_{ij}]$ is an element at position (i, j) which is i^{th} row and j^{th} column in matrix A, and
- $[a_{ji}]$ is an element at position (j, i) which is j^{th} row and i^{th} column in matrix A.

For Example, consider $A = \begin{pmatrix} 0 & 3 & -8 \\ -3 & 0 & 6 \\ 8 & -6 & 0 \end{pmatrix}$,

and *Transpose of A* $= \begin{pmatrix} 0 & -3 & 8 \\ 3 & 0 & -6 \\ -8 & 6 & 0 \end{pmatrix}$

Here, in the transposed form the matrix looks like the negative of the original matrix.

Properties of Skew-Symmetric Matrices

There are some rules that come from the concept of Symmetric and Skew-Symmetric Matrices,

- All eigenvalues of a skew-symmetric matrix are either 0 or purely imaginary.
- Eigenvectors corresponding to distinct non-zero eigenvalues of a skew-symmetric matrix are orthogonal.
- The determinant of a skew-symmetric matrix is 0 if the size of the matrix is odd, and the determinant is the product of the eigenvalues if the size of the matrix is even.
- Skew-symmetric matrices with complex entries are called skew-hermitian matrices, here instead of transpose we take the conjugate transpose of the matrix.

Property: If matrix A is a square matrix then $(A - A^t)$ is always skew-symmetric.

Proof:

To find if a matrix skew-symmetric or not, first, we have to find the transposed form of the given matrix

So, let's find the transpose of $(A - A^t)$

$$= (A - A^t)^t$$

$$= A^t - (A^t)^t$$

$$= A^t - A \text{ [here, } (A^t)^t = A]$$

$$= -(A - A^t)$$

So, this form is the negative of the given matrix, so it is skew-symmetric.

Determinant of Skew Symmetric Matrix

Skew-Symmetric Matrix, is a square matrix and the [determinant](#) of the Skew-Symmetric matrices follows the condition discussed below. If we have a Skew-Symmetric Matrix then,

$$\det(A^T) = \det(-A) = (-1)^n \det(A)$$

Also, every skew-symmetric matrix of odd order is a **singular matrix**, i.e. its determinant is zero and hence its determinant does not exist.

Expressing Matrix in the form of Symmetric and Skew-Symmetric Matrices

Property: Every square matrix can be expressed uniquely as the sum of symmetric and skew-symmetric matrices.

Proof:

Let A be a square matrix,

We can write, $A = \frac{A+A^T}{2} + \frac{A-A^T}{2}$

$$= \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T) = \frac{1}{2}(A+A^T) + \frac{1}{2}(A-A^T)$$

Let, $A = P + Q$

Where, $P = \frac{1}{2}(A+A^T)$, $Q = \frac{1}{2}(A-A^T)$ $P = \frac{1}{2}(A+A^T)$, $Q = \frac{1}{2}(A-A^T)$

Now, find P^T and Q^T [P^T is the **transpose** of P]

$$P^T = (\frac{1}{2}(A+A^T))^T = \frac{1}{2}(A+A^T)^T = \frac{1}{2}(A^T+(A^T)^T) = \frac{1}{2}(A+A^T) = P \quad P^T = (\frac{1}{2}(A+A^T))^T = \frac{1}{2}(A^T+(A^T)^T) = \frac{1}{2}(A+A^T) = P$$

and,

$$Q^T = (\frac{1}{2}(A-A^T))^T = \frac{1}{2}(A-A^T)^T = \frac{1}{2}(A^T-(A^T)^T) = \frac{1}{2}(A^T-A) = -\frac{1}{2}(A-A^T) = -Q \quad Q^T = (\frac{1}{2}(A-A^T))^T = \frac{1}{2}(A^T-(A^T)^T) = \frac{1}{2}(A^T-A) = -\frac{1}{2}(A-A^T) = -Q$$

So, here **P** is symmetric and **Q** is skew-symmetric matrices and **A** is the sum of **P** and **Q**.

Matrix A :

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Step 1: Transpose of A , denoted A^T :

$$A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

Step 2: Symmetric part S :

The symmetric part S is calculated as: $S = \frac{1}{2}(A + A^T)$

$$\text{Adding } A \text{ and } A^T: A + A^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} + \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 2 & 6 & 10 \\ 6 & 10 & 14 \\ 10 & 14 & 18 \end{bmatrix}$$

$$\text{Now divide by 2: } S = \frac{1}{2} \begin{bmatrix} 2 & 6 & 10 \\ 6 & 10 & 14 \\ 10 & 14 & 18 \end{bmatrix} = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 7 \\ 5 & 7 & 9 \end{bmatrix}$$

Step 3: Skew-symmetric part K :

The skew-symmetric part K is calculated as: $K = \frac{1}{2}(A - A^T)$

$$\text{Subtract } A^T \text{ from } A: A - A^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} = \begin{bmatrix} 0 & -2 & -4 \\ 2 & 0 & -2 \\ 4 & 2 & 0 \end{bmatrix}$$

$$\text{Now divide by 2: } K = \frac{1}{2} \begin{bmatrix} 0 & -2 & -4 \\ 2 & 0 & -2 \\ 4 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}$$

Final Result:

$$\text{Thus, we can express } A \text{ as: } A = S + K \text{ where: } S = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 7 \\ 5 & 7 & 9 \end{bmatrix}, \quad K = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}$$

Unitary Matrix

A square matrix of complex numbers is said to be a unitary matrix if its inverse is equal to the conjugate transpose.

- $UU^H = U^H U = I$
- $U^H = U^{-1}$

Unitary Matrix Definition

A unitary matrix is a square matrix that has some properties analogous to those of orthogonal matrices but for complex numbers.

Unitary Matrix Examples

- Matrix given below is a unitary matrix of order “2 × 2.”

$$A_{2 \times 2} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ i & -i \end{bmatrix}$$

Matrix given below is a unitary matrix of order “3 × 3.”

$$B_{3 \times 3} = \begin{bmatrix} i & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & i \end{bmatrix}$$

Normal Matrix

A square matrix A of order $n \times n$ of complex numbers, that is $A \in \mathbb{C}^{n \times n}$ is said to be a normal matrix if $A^H A = A A^H$ where A^H is the conjugate transpose of A .

The matrix $A \in \mathbb{C}^{n \times n}$ is a Normal Matrix $\Leftrightarrow A^H A = A A^H$

Properties of Normal Matrix

There are a few interesting properties of a Normal matrix are listed below which are useful in deriving many more results in Linear Algebra.

- If A is a normal matrix then it is diagonalizable by a unitary matrix.
- A Hermitian matrix is a normal matrix.

Let A be a Hermitian matrix then $A^H = A$, then

$$\text{Now, } A^H A - A A^H = A A - A A = A^2 - A^2 = 0$$

$$\Rightarrow A^H A - A A^H = 0 \Rightarrow A^H A = A A^H$$

$\Rightarrow A$ is normal

- A skew-Hermitian matrix is a normal matrix.
- A unitary matrix is normal.

If U is a unitary matrix, then $U^H U = U U^H = I$, hence normal.

- A symmetric and a skew-symmetric matrix both are normal matrices.
- A normal matrix need not be a Hermitian, skew-Hermitian, Unitary or symmetric matrix.
- An orthogonal matrix is also a normal matrix.
- If A is normal then, AA^H is a Hermitian matrix.
- If A is normal, then there exists a set of orthonormal eigenvectors of A in C^n .
- A normal matrix is unitary if and only if its eigenvalues lies on the unit circle in complex plane.

Examples of Normal Matrix

Let us take an example of a complex matrix of order 2×2 ,

$$\begin{bmatrix} 4i & -1+i \\ 1-i & 4i \end{bmatrix}$$

So according to the definition of normal matrix let us find the conjugate transpose of the given matrix.

Let

$$A = \begin{bmatrix} 4i & -1+i \\ 1-i & 4i \end{bmatrix}$$

Then, $A^H = (\text{conjugate of } A)^T =$

$$\begin{aligned} & \begin{bmatrix} -4i & -1-i \\ 1+i & -4i \end{bmatrix}^T \\ &= \begin{bmatrix} -4i & 1+i \\ -1-i & -4i \end{bmatrix} \end{aligned}$$

Then $A^H A =$

$$\begin{bmatrix} -4i & 1+i \\ -1-i & -4i \end{bmatrix} \begin{bmatrix} 4i & -1+i \\ 1-i & 4i \end{bmatrix} \\ = \begin{bmatrix} 18 & 8i \\ -8i & 18 \end{bmatrix}$$

And $AA^H =$

$$\begin{bmatrix} 4i & -1+i \\ 1-i & 4i \end{bmatrix} \begin{bmatrix} -4i & 1+i \\ -1-i & -4i \end{bmatrix} \\ = \begin{bmatrix} 18 & 8i \\ -8i & 18 \end{bmatrix}$$

Hence $A^H A = AA^H$, that is a normal matrix.

Let us take another example of a real matrix and check whether it satisfies the condition of a normal matrix.

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

,

For a real matrix, the conjugate transpose will be the same as the transpose of matrix.

Therefore, $A^H = A^T =$

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

Now, $A^T A =$

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

And $AA^T =$

$$\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

Thus, $A^T A = AA^T$, A is normal.

With this above example we can also say that orthogonal matrices are also normal matrix.

Idempotent Matrix

A square matrix “P” of any order, and the matrix P is said to be an idempotent matrix if and only if $P^2 = P$. Idempotent matrices are singular and can have non-zero entries. Every identity matrix is also an idempotent matrix, as the identity matrix gives the same matrix when multiplied by itself.

Examples of Idempotent Matrix

The matrix given below is an idempotent matrix of order “2 × 2.”

$$A_{2,2} = \begin{bmatrix} 4 & -1 \\ 12 & -3 \end{bmatrix}$$

Properties of Idempotent Matrix

The following are some important properties of an idempotent matrix:

- Every idempotent matrix is a square matrix.
- All idempotent matrices are singular matrices, apart from the identity matrix.
- The determinant of an idempotent matrix is either one or zero.
- The non-diagonal entries of an idempotent matrix can be non-zero entries.
- The trace of an idempotent matrix is always an integer and equal to the rank of the matrix.
- The eigenvalues of an idempotent matrix are either zero or one.
- The following is the relationship between idempotent and involuntary matrices: A square matrix “A” is said to be an idempotent matrix if and only if $P = 2A - I$ is an involuntary matrix.

Involuntary Matrix

A square matrix is said to be an involuntary matrix that, when multiplied by itself, gives an identity matrix of the same order. A square matrix “P” is said to be an involuntary matrix if its inverse is the original matrix itself i.e. $P = P^{-1}$

Or $A^2 = I$.

Properties of involutory Matrix

The following are some important properties of an involutory matrix:

- A square matrix “A” of any order is said to be involutory if and only if $A^2 = I$ or $A = A^{-1}$.
- If A and B are two involutory matrices of the same order and $AB = BA$, then AB is also an involutory matrix.
- The determinant of an involutory matrix is always either -1 or +1.
- If “A” is an involutory matrix of any order, then $A^n = I$ if n is even and $A^n = A$ if n is odd, where n is an integer.
- If a block diagonal matrix is derived from an involutory matrix, then the obtained matrix is also involutory.
- The eigenvalues of an involutory matrix are always either -1 or +1.
- Symmetric involutory matrix is orthogonal, and vice versa.
- An involutory matrix “A” can also be an idempotent matrix if “A” is an identity matrix.
- The following is the relationship between idempotent and involutory matrices: A square matrix “A” is said to be an involutory matrix if and only if $A = \frac{1}{2}(B + I)$, where B is an idempotent matrix.

Nilpotent Matrix

A square matrix “A” of order “ $n \times n$ ” is said to be nilpotent if “ $A^k = O$,” where “O” is a null matrix of order “ $n \times n$ ” and “k” is a positive integer less than or equal to n.

$A_{n \times n}$ is a Nilpotent Matrix,

$$\text{If } A^k = O$$

$$k \leq n$$

Rank of Matrix

Rank of a matrix is defined as the number of linearly independent rows in a matrix.

It is denoted using $\rho(A)$ where A is any matrix. Thus the number of rows of a matrix is a limit on the rank of the matrix, which means the rank of the matrix cannot exceed the total number of rows in a matrix.

For example, if a matrix is of the order 3×3 then the maximum rank of a matrix can be 3.

Note: If a matrix has all rows with zero elements, then the rank of a matrix is said to be zero.

How To Calculate Rank of a Matrix?

There are 3 methods which can be used to get the rank of any given matrix. These methods are as follows:

- Minor Method
- Using Echelon Form
- Using Normal Form

Minor Method

In order to find the rank of a matrix using minor method, following steps are followed:

- Calculate the determinant of the matrix (say A). If $\det(A) \neq 0$, then rank of matrix A = order of matrix A .
- If $\det(A) = 0$, then the rank of the matrix is equal to order of the maximum possible non zero minor of the matrix.

Let us understand how to find the rank of matrix using minor method.

Example: Find the rank of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 7 \end{bmatrix}$ using minor method.

Given $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 7 \end{bmatrix}$

- Step 1: Calculate the determinant of A

$$\det(A) = 1(35 - 48) - 2(28 - 42) + 3(32 - 35)$$

$$\det(A) = -13 + 28 - 9 = 6$$

- As $\det(A) \neq 0$, $\rho(A) = \text{order of } A = 3$

Using Echelon Form

The minor method becomes very tedious if the order of the matrix is very large. So in this case, we convert the matrix into Echelon Form. A matrix that is in upper triangular form or lower triangular form is considered to be in Echelon Form. A matrix can be converted to its Echelon Form by using elementary row operations. Following steps are followed to calculate the rank of a matrix using Echelon form:

- Convert the given matrix into its Echelon Form.
- The number of non-zero rows obtained in the Echelon form of the matrix is the rank of the matrix.

Let us understand how to find the rank of matrix using minor method.

Example: Find the rank of matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ using Echelon form method.

Given $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$

- Step 1: Convert A to echelon form

$$\text{Apply } R_2 = R_2 - 4R_1$$

$$\text{Apply } R_3 = R_3 - 7R_1$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{bmatrix}$$

$$\text{Apply } R_3 = R_3 - 2R_2$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{bmatrix}$$

As matrix A is now in lower triangular form, it is in Echelon Form.

- Step 2: Number of non-zero rows in $A = 2$. Thus $\rho(A) = 2$

Using Normal Form

A matrix is said to be in normal form if it can be reduced to the form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$.

Here I_r represents the identity matrix of order r . If a matrix can be converted to its normal form, then rank of the matrix is said to be r .

Let us understand how to find the rank of matrix using minor method.

Example: Find the rank of matrix $\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$ using normal form method.

$$\text{Given } A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$$

Apply $R_2 = R_2 - R_1$, $R_3 = R_3 - 2R_1$ and $R_4 = R_4 - 3R_1$

$$A = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Apply $R_1 = R_1 - 2R_2$ and $R_4 = R_4 - R_2$

$$A = \begin{bmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Apply $R_1 = R_1 + R_3$ and $R_2 = R_2 - R_3$

$$A = \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Apply $C_4 \rightarrow C_4 - 2C_1$

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Thus A can be written as $\begin{bmatrix} I_3 & 0 \\ 0 & 0 \end{bmatrix}$.

Thus, $\rho(A) = 3$

Properties of Rank of Matrix

Properties of rank of matrix is as follows:

- Rank of a matrix is equal to the order of the matrix if it is a non-singular matrix.
- Rank of a matrix is equal to the number of non-zero rows if it is in Echelon Form.
- Rank of matrix is equal to the order of identity matrix in it if it is in normal form.
- Rank of matrix < Order of matrix if it is singular matrix.
- Rank of matrix < minimum {m, n} if it is a rectangular matrix of order m x n.
- Rank of identity matrix is equal to the order of the identity matrix.
- Rank of a zero matrix or a null matrix is zero.

Echelon Form of a Matrix

Echelon form of a matrix is used to solve linear equations. It is effective in changing a complex matrix into a simple matrix. A matrix is in echelon form if it satisfies some conditions which we are discussed below:

- The first non zero element in a row is 1. This entry is known as pivot.
- Each pivot in a column is in the right side of the pivot column in a previous row.
- A zero row should be below the non zero rows.

Reduced Row-Echelon Form 

$$\begin{bmatrix} 1 & 3 & 3 & 8 & 5 \\ 0 & 1 & 3 & 10 & 8 \\ 0 & 0 & 0 & -1 & -4 \\ 0 & 0 & 0 & 2 & 8 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 & -6 & 0 & 64 \\ 0 & 1 & 3 & 0 & -32 \\ 0 & 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

This lesson introduces the concept of an **echelon matrix**. Echelon matrices come in two forms: the **row echelon form** (ref) and the **reduced row echelon form** (rref).

Row Echelon Form

A matrix is in **row echelon form** (ref) when it satisfies the following conditions.

- The first non-zero element in each row, called the **leading entry**, is 1.
- Each leading entry is in a column to the right of the leading entry in the previous row.
- Rows with all zero elements, if any, are below rows having a non-zero element.

Each of the matrices shown below are examples of matrices in row echelon form.

$$\begin{array}{ccc} \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} & \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \\ \mathbf{A_{ref}} & \mathbf{B_{ref}} & \mathbf{C_{ref}} \end{array}$$

Note: Some references present a slightly different description of the row echelon form. They do not require that the first non-zero entry in each row is equal to 1.

Reduced Row Echelon Form

A matrix is in **reduced row echelon form** (ref) when it satisfies the following conditions.

- The matrix is in row echelon form (i.e., it satisfies the three conditions listed above).
- The leading entry in each row is the only non-zero entry in its column.

Each of the matrices shown below are examples of matrices in *reduced* row echelon form

$$\begin{array}{ccc}
 \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} &
 \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} &
 \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \\
 \mathbf{A_{ref}} & \mathbf{B_{ref}} & \mathbf{C_{ref}}
 \end{array}$$

Steps to Convert a Matrices to Echelon Form

1. **Identify the Leading Entry:** Start with the first column and find the topmost non-zero entry. This will be your leading entry for the first row.
2. **Row Swaps:** If necessary, swap rows to position the leading entry at the top.
3. **Eliminate Below:** Use row operations (adding or subtracting multiples of the leading row from the rows below) to create zeros below the leading entry.
4. **Move to the Next Column:** Move to the next column to the right and repeat steps 1-3 for the submatrix that remains after the current leading row.
5. **Repeat Until Complete:** Continue this process until you've processed all columns.

Row canonical form

A matrix A is said to be in **row canonical form** (or **reduced row echelon form** (RREF)) if the following conditions hold (where a *leading nonzero element* of a row of A is the first nonzero element in the row):

1. **All Zero Rows at the Bottom:** If there are any rows that contain all zeros, those rows should be at the bottom of the matrix.
2. **Leading Entries to the Right:** Each row with a non-zero entry should have its leading entry (the first non-zero number from the left) to the right of the leading entry of the row above it.
3. **Leading Entry is 1:** Every leading entry (pivot) must be equal to 1.
4. **Only Non-Zero Entry in Its Column:** Each leading entry must be the only non-zero number in its column.

Linear Dependence, Linear Independence

1. Linear Independence:

A set of vectors $\{v_1, v_2, \dots, v_n\}$ is **linearly independent** if the only solution to the equation

$$c_1v_1 + c_2v_2 + \dots + c_nv_n = 0$$

is $c_1 = c_2 = \dots = c_n = 0$.

2. Linear Dependence:

A set of vectors is **linearly dependent** if there exist coefficients $c_1, c_2, \dots, c_n$, not all zero, such that

$$c_1v_1 + c_2v_2 + \dots + c_nv_n = 0.$$

Methods to Check Linear Dependence/Independence

Method 1: Row Reduction

1. Form a Matrix:

Arrange the vectors as columns in a matrix A .

2. Perform Row Reduction:

Use Gaussian elimination to bring A to row echelon form or reduced row echelon form.

3. Analyze Pivot Columns:

- If every column has a pivot (leading 1), the vectors are linearly independent.
- If any column lacks a pivot, the vectors are linearly dependent.

Method 2: Determinant Test (*For Square Matrices Only*)

1. Ensure the Matrix is Square:

The number of vectors must equal the number of dimensions (rows = columns).

2. Compute the Determinant:

- If $\det(A)=0$ the vectors are linearly dependent.
- If $\det(A)\neq 0$, the vectors are linearly independent.

Example: Vectors $(1, 2)$, $(2, 4)$, and $(3, 6)$

Step-by-Step Using Row Reduction:

1. **Form a Matrix:**

Arrange the vectors as columns:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

2. **Row Reduce:**

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix} \xrightarrow{R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

3. **Check Pivot Columns:**

- The first column has a pivot (1).
- The second row is all zeros, indicating not all columns have pivots.

Conclusion:

The vectors are linearly dependent because the row-reduced matrix has a row of zeros.

System of Linear Equation:

- **Consistent System:** A system of equations that has at least one solution.
- **Inconsistent System:** A system of equations that has no solutions.